

Digital Education in Sri Lanka: A Bibliometric Analysis

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INTRODUCTION

- Digital education has transformed teaching and learning worldwide.
- ICT has accelerated digital learning in schools and higher education.
- Sri Lanka has introduced digital education policies to improve access, infrastructure and learning opportunities.
- Understanding research trends is important for future policy and educational development.
- Bibliometric analysis provides evidence on how digital education research has evolved in Sri Lanka.





SIGNIFICANCE OF THE STUDY

- Identifies publication trends in digital education research
- Recognizes leading researchers and journals
- Highlights research gaps and emerging themes
- Supports evidence-based educational policymaking
- Guides future researchers and funding agencies
- Provides the first comprehensive bibliometric overview of digital education research in Sri Lanka using the OpenAlex database

OBJECTIVES

- To identify the patterns and extend of growth of Scientific Literature published in Sri Lanka from 2015-2026 and as indexed in the OpenAlex database
- To identify the various types of literature on digital education in Sri Lanka, published from 2015 to 2026
- To identify the most influential authors and sources contributing to digital education in Sri Lanka
- What keywords are frequently used by authors in digital education in Sri Lanka
- To study the correlation between the h-index of authors and journals.
- To provide recommendations for future research related to digital education in Sri Lanka

METHODOLOGY

- Primary data was obtained from the **OpenAlex database**.

"OpenAlex is a nonprofit, open catalogue of the global research system, (Simard, M.A. et al., 2024) and is rapidly gaining popularity. Also it combines access to records from databases such as PubMed, Web of Science, Scopus, and the Cochrane Library databases, while allowing the exclusion of repetitive records from the analysis (Culbert, J. et al., 2024)"

- OpenAlex database were used to retrieve peer-reviewed articles on Digital Education in Sri Lanka, published in year 2015–2026.

- Search in titles and abstracts : **"Digital Education" AND "Sri Lanka". Instead of digital education the terms "digital learning" OR "e-learning" OR "online learning" OR "distance learning" OR "virtual learning" OR "ICT in education"** and found 5,848 articles.

METHODOLOGY

- **Inclusion criteria:** Scholarly publications such as articles in peer-reviewed journals on digital education in Sri Lanka
- **Exclusion criteria:** Scholarly papers are not in English, journal articles that have not been peer-reviewed and papers in unrelated areas or that are irrelevant to the specific subject of the title
- 459 articles were selected
- Exported to **Biblioshiny**

(*it is a free bibliometric tool with integration that allows users to receive automatic interpretations, critical insights, and narrative summaries of their bibliometric results directly within the platform)*

- Quantitative methods were used to analyze the data and explain the results

Bibliometric Overview



- 459 publications
- 200 journals/sources
- 1,095 authors , Only 94 authors published single authored research articles.
- 731 author keywords
- 5,447 cited references
- Notable finding : annual growth rate of -15.65% which indicates overall decline in research output over the study period.
- Digital education has become an important multidisciplinary research area in Sri Lanka.

Annual Scientific Production

- Publications steadily increased from 2015
- Peak research output:
  - **2021 – 76 publications**
- Declining trend after 2021

- **Highest Citation Impact**

  2019 publications

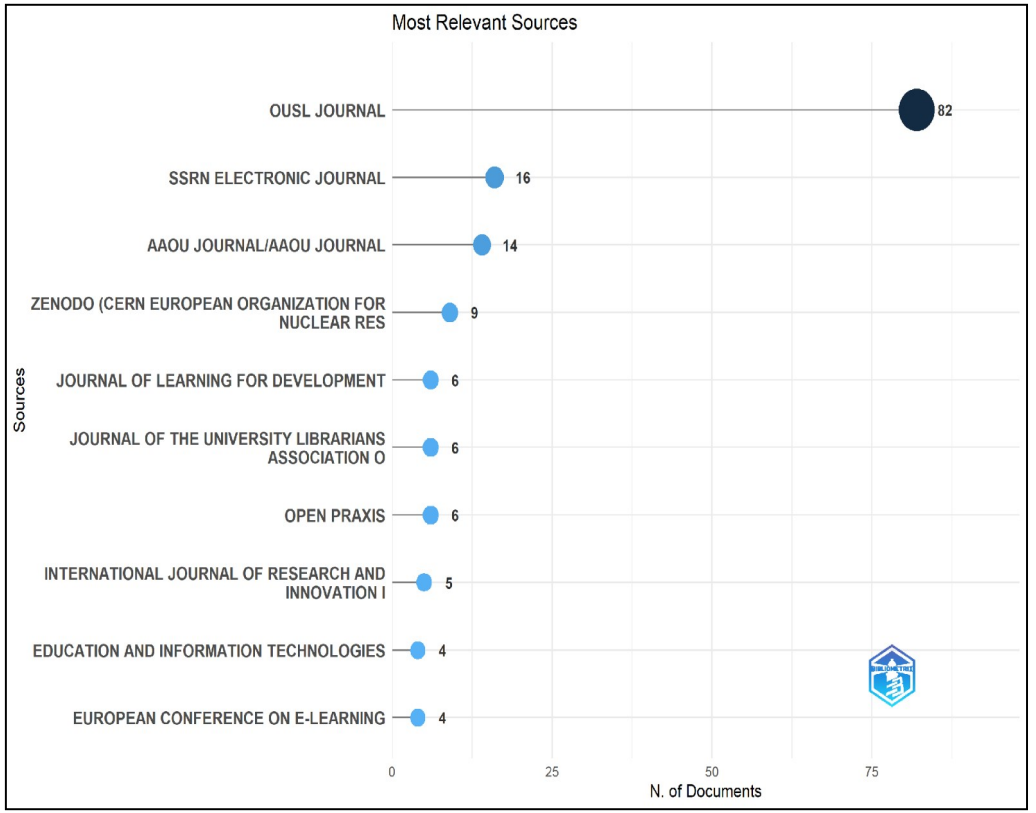
  Mean citations = **16.88/article**

Year	Articles	MeanTCperArt	MeanTCperYear
2015	26	4.19	0.35
2016	31	5.35	0.49
2017	30	4.93	0.49
2018	22	4.18	0.46
2019	34	16.88	2.11
2020	45	11.73	1.68
2021	76	7.25	1.21
2022	66	4.62	0.92
2023	52	3.29	0.82
2024	36	1.17	0.39
2025	37	0.41	0.2
2026	4	0.25	0.25

Leading Sources & Influential Authors

Top Source

- OUSL Journal (82 publications)
- Other leading sources
- SSRN Electronic Journal
- AAOU Journal
- Zenodo
- Journal of Learning for Development



Most Influential Author

- **Shironica P. Karunanayaka**
- H-index = 6

Research Themes

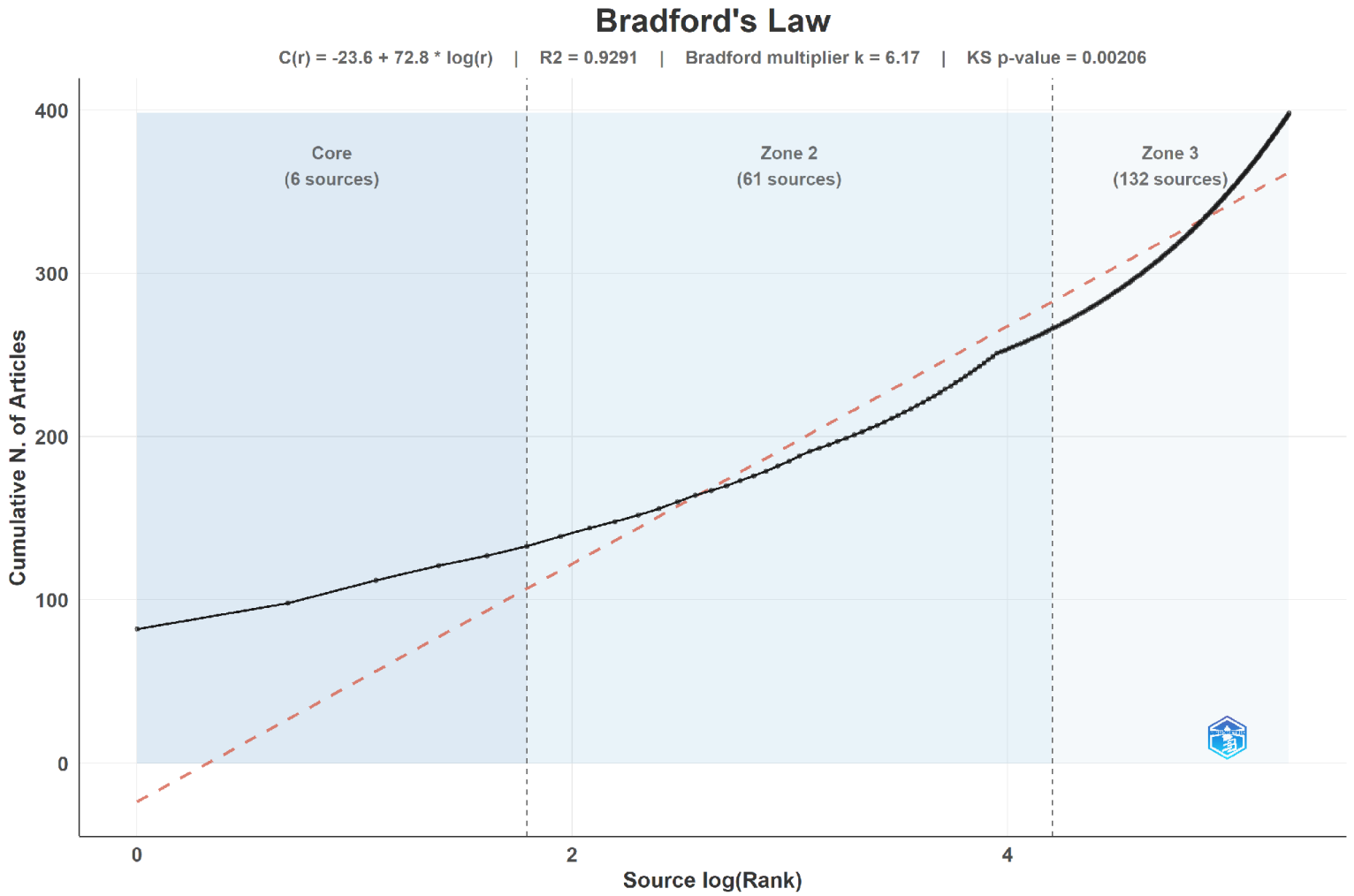
Most Frequent Terms

- Computer Science
- Sri Lanka
- Psychology
- Sociology
- Medical Education



- Digital education research is becoming increasingly interdisciplinary.

Bradford’s Law



DISCUSSION

- COVID-19 significantly accelerated digital education research.
- Emergency online learning increased publications during 2020–2021.
- Research declined after 2021 due to
  - Return to face-to-face learning
  - Sri Lanka's economic crisis
  - Reduced research funding
- Future research should focus more on pedagogy than technology alone.

CONCLUSION

- Digital education research has expanded considerably in Sri Lanka.
- Bibliometric analysis identified publication trends, leading journals, influential authors and emerging themes.
- Findings support evidence-based educational planning.
- Greater emphasis is needed on:
  - Digital literacy
  - AI literacy
  - Data literacy
- Competency-based curricula
- The study provides a roadmap for future digital education research and policy development.

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Thank you for your attention.